On Improvements of Simplified and Highly Stable Lattice Boltzmann Method: Formulations, Boundary Treatment, and Stability Analysis

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Abstract

In this paper, we present a detailed report on a revised form of simplified and highly stable lattice Boltzmann method (SHSLBM) and its boundary treatment as well as stability analysis. SHSLBM is a recently developed scheme within lattice Boltzmann (LB) framework, which utilizes lattice properties and relationships given by Chapman-Enskog (C-E) expansion analysis to reconstruct solutions of macroscopic governing equations recovered from lattice Boltzmann equation (LBE) and resolved in a predictor-corrector scheme. Formulations of original SHSLBM are slightly adjusted in the present work to facilitate implementation on body-fitted mesh. The boundary treatment proposed in this paper offers an analytical approach to interpret no-slip boundary condition. And the stability analysis in this paper fixes flaws in previous works and reveals very nice stability characteristic in high Reynolds number scenarios. Several benchmark tests are conducted for comprehensive evaluation of

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This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1002/fld.4485

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the boundary treatment and numerical validation of stability analysis. It turns out that by adopting the modifications suggested in this work, lower numerical error can be expected.

Key words: lattice Boltzmann method, stability analysis, boundary treatment, Chapman-Enskog expansion analysis, incompressible flow, viscous flows

1 Introduction

Lattice Boltzmann Method (LBM) is an attractive Computational Fluid Dynamics (CFD) method due to its kinetic nature, simplicity, and explicitness [1-10]. It offers a special perspective in studying fluid mechanics problems, in which the evolution of the distribution functions is tracked during the computation. And the macroscopic variables and governing equations can be obtained from LBM solutions by conservation laws and Chapman-Enskog expansion analysis [11]. The two processes in LBM, namely steaming and collision, only involve a set of simple algebraic equations, and are able to reflect complex nonlinear physical phenomenon. Apart from its various merits, some drawbacks exist in conventional LBM, which include more cost in virtual memories, inconvenient implementation of physical boundary conditions and poor numerical stability at high Reynolds number [12-19].

To overcome these drawbacks, a Simplified Lattice Boltzmann Method (SLBM) was recently developed [20, 21]. In SLBM, the macroscopic equations recovered by Lattice Boltzmann Equation (LBE) are resolved by a predictor-corrector scheme, which are then reconstructed by the lattice properties and the relationships given by Chapman-Enskog (C-E) expansion analysis; and the final formulations only involve the equilibrium and the non-equilibrium distribution functions. In its computational process, the equilibrium distribution functions are calculated from the macroscopic variables, while the non-equilibrium distribution functions are evaluated from the difference of two equilibrium distribution

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functions at different locations and time levels. As a consequence, SLBM directly tracks the evolution of the macroscopic variables rather than that of the distribution functions, which remarkably saves the virtual memories and facilitates the implementation of physical boundary conditions. Note that SLBM is different from the lattice kinetic scheme [22, 23] proposed in previous studies although the formulation of lattice kinetic scheme is similar to the formulation of SLBM in the predictor step. The lattice kinetic scheme introduces a different form of distribution function which consolidates spatial derivative terms. Such strategy may bring some numerical diffusions into computation.

However, it is noteworthy that flaws exist in previous stability analysis of SLBM, although its stability performance was well revealed in numerical tests [21]. Therefore, the previous claim of "unconditionally stable" is not true. In this paper, we present a robust von Neumann stability analysis on the method. Further comparison with similar analysis performed on conventional LBM [5] indicates that the present method is much more stable than conventional LBM in high Reynolds number scenarios. Thus, instead of claiming it as an unconditionally stable scheme, here we rename the method as simplified and highly stable lattice Boltzmann method (SHSLBM). Another notable issue is the implementation of no-slip boundary condition. In the previous work, boundary condition for non-equilibrium distribution function is still required, for which the non-equilibrium extrapolation scheme is adopted [21]. This boundary treatment is based on the assumption that the field of distribution functions is smooth. So basically, it is a mathematical approach to implement boundary conditions. In this paper, we offer an analytical interpretation of the no-slip boundary condition. No extrapolation scheme for the non-equilibrium term is required, due to which the present boundary treatment is physically more robust. Furthermore, the formulations of original SLBM are also revised to make the present scheme more robust and facilitate its implementation on body-fitted non-uniform mesh.

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The remaining parts of this paper are organized as follows. Section 2 presents derivations and boundary treatments of simplified and highly stable lattice Boltzmann method (SHSLBM). Von Neumann stability analysis is performed in Section 3. In Section 4, five benchmark tests are conducted to demonstrate the accuracy, the robustness, and the flexibility of the new method. Finally, some conclusions are drawn in Section 5.

2 Simplified and Highly Stable Lattice Boltzmann Method (SHSLBM)

2.1 Original formulations

The simplified and highly stable lattice Boltzmann method (SHSLBM), or previously named USLBM [21], is derived by reconstructing solutions to macroscopic equations recovered from lattice Boltzmann equation and resolved in a predictor-corrector approach. Specifically, formulations of SHSLBM are

Predictor step:

$$\rho^* = \sum_\alpha f_\alpha^{eq}\left(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t\right) \tag{1}$$

$$\rho^*\mathbf{u}^* = \sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\left(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t\right) \tag{2}$$

Corrector Step:

$$\rho(\mathbf{r},t) = \rho^* \tag{3}$$

$$\rho(\mathbf{r},t)\mathbf{u}(\mathbf{r},t) = \rho^* u^* - \left(1-\frac{1}{\tau}\right)\sum_\alpha \mathbf{e}_\alpha\left[f_\alpha^{neq}\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) - f_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right)\right] \tag{4}$$

where $\mathbf{r}$ and t are space location and time level, respectively; ρ, $\mathbf{u}$, $\mathbf{e}_\alpha$ and δ_t represent density, velocity, lattice velocity vector, and time interval, respectively. τ denotes the single relaxation parameter in LBGK model, which is related to the kinematic viscosity υ by

$$\upsilon = c_s^2\delta_t\left(\tau-0.5\right) \tag{5}$$

The equilibrium distribution function is calculated from

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$$f_\alpha^{eq} = \rho\omega_\alpha\left[1+\frac{\mathbf{e}_\alpha\cdot\mathbf{u}}{c_s^2}+\frac{\left(\mathbf{e}_\alpha\cdot\mathbf{u}\right)^2-\left(c_s|\mathbf{u}|\right)^2}{2c_s^4}\right] \tag{6}$$

where ω_α denotes the weighting parameter along the α-direction in lattice velocity model; c_s the sound speed. And the non-equilibrium distribution function is evaluated by

$$f_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t,t-\frac{1}{2}\delta_t\right)=-\tau\left[f_\alpha^{eq}(\boldsymbol{r},t)-f_\alpha^{eq}\left(\mathbf{r}-\mathbf{e}_\alpha\delta_t,t-\delta_t\right)\right] \tag{7}$$

Referring to the work of Shu et al [24], intermediate macroscopic variables are used to evaluate the first equilibrium distribution function on the RHS of Eq. (7) defined at the time level *t*.

As can be seen, the final formulations of SHSLBM only involve equilibrium and non-equilibrium distribution functions. In practical computations, the equilibrium distribution function is calculated from the macroscopic fluid properties as shown in Eq. (6), while the non-equilibrium term is evaluated by adopting Eq. (7) that uses two equilibrium distribution functions at different locations and time levels. Consequently, SHSLBM practically tracks the evolution of macroscopic fluid properties, rather than the distribution functions. Lower cost in virtual memories can then be expected, since it is no longer necessary to store the distribution functions. Apart from that, the present method also facilitates the implementation of physical boundary condition, and has better stability performance as compared to the conventional LBM, which will be discussed in greater detail in the following sections. The above scheme can be flexibly extended to non-uniform meshes by consolidating Lagrange interpolation algorithms.

2.2 Revised formulations of SHSLBM

The formulations of the corrector step reported in previous work can be further derived to facilitate implementation of SHSLBM on non-uniform body-fitted meshes. This is realized

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by simply substituting the approximation formulation of non-equilibrium distribution function (7) into the corrector step (Eq. (4)), which gives

$$\rho(\mathbf{r},t)\mathbf{u}(\mathbf{r},t)=\rho^{*}\mathbf{u}^{*}+(\tau-1)\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{eq}(\mathbf{r}+\mathbf{e}_{\alpha}\delta_{t},t)-(\tau-1)\rho(\mathbf{r},t-\delta_{t})\mathbf{u}(\mathbf{r},t-\delta_{t}) \tag{8}$$

Note that equilibrium distribution function in the above formulation is evaluated from the intermediate variables obtained in the predictor step, and Eq. (2) and the following relationship are used in deriving the above formulation

$$\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_{\alpha}\delta_{t},t-\frac{1}{2}\delta_{t}\right)=-\tau\sum_{\alpha}\mathbf{e}_{\alpha}\left[f_{\alpha}^{eq}(\boldsymbol{r},t)-f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_{t},t-\delta_{t})\right]=0 \tag{9}$$

Two advantages can be expected by adopting Eq. (8) instead of Eq. (4), especially when body-fitted non-uniform mesh is adopted. Firstly, on body-fitted mesh, the projection from the physical coordinate to the computational coordinate is no longer linear (see **Figure 1**). Therefore, the shifting of physical coordinate from $\mathbf{r}$ to ($\mathbf{r}$-0.5$\mathbf{e}_{\alpha}\delta_{t}$) does not linearly correspond to a similar shifting in the computational coordinate (along η- and ξ- directions). So the computed non-equilibrium distribution function as shown in Eq. (7) does not locate on the node of computational coordinate, due to which adopting Lagrange interpolation will inevitably introduce errors. This issue can be resolved either by using other interpolation algorithms which are valid on arbitrary distributed nodes or by simply applying Eq. (8) in the corrector step. Obviously, the latter strategy is relatively easier in implementation. The known values of equilibrium distribution functions are all on the computational nodes. And due to the symmetricity of lattice velocity vectors, the interpolation points are the same as those used in the predictor step. Therefore, Lagrange interpolation can still be applied and the interpolation parameters in the predictor step can still be adopted in the corrector step. The other advantage is reflected in the boundary treatment. By using Eq. (8), the boundary condition for the non-equilibrium distribution function is no longer needed. Because the

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equilibrium distribution function can be calculated from the macroscopic variables, the physical boundary condition can be directly implemented in SHSLBM.

The final formulations of the revised scheme of SHSLBM read:

Predictor step:

$$\rho^* = \sum_\alpha f_\alpha^{eq}\left(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t\right) \tag{10}$$

$$\rho^*\mathbf{u}^* = \sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\left(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t\right) \tag{11}$$

Corrector Step:

$$\rho(\mathbf{r},t) = \rho^* \tag{12}$$

$$\rho(\mathbf{r},t)\mathbf{u}(\mathbf{r},t) = \rho^*\mathbf{u}^* + (\tau-1)\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\left(\mathbf{r}+\mathbf{e}_\alpha\delta_t, t\right) - (\tau-1)\rho(\mathbf{r},t-\delta_t)\mathbf{u}(\mathbf{r},t-\delta_t) \tag{13}$$

Note that the approximation of non-equilibrium term has already been incorporated into the formulation, and the final expressions only involve the equilibrium terms and the macroscopic variables.

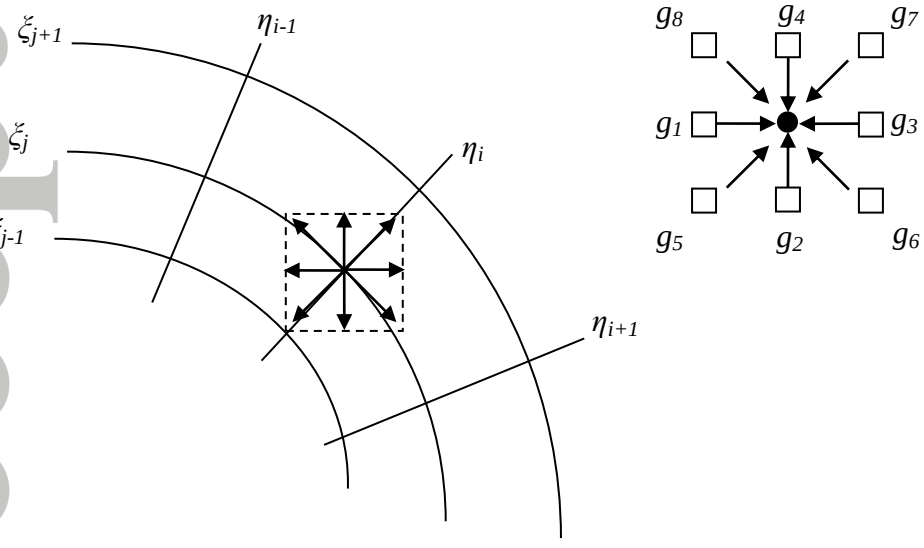


Figure 1 Illustration of constructing D2Q9 model in a body-fitted mesh

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2.3 Implementation of Boundary Conditions

One widely-recognized drawback of conventional LBM is its inconvenient implementation of physical boundary conditions. Since LBM directly updates the distribution functions, the physical boundary conditions about the macroscopic fluid properties need to be transformed into the boundary conditions for the distribution functions so as to be implemented in LBM [1].

The present simplified and highly stable lattice Boltzmann method (SHSLBM) offers an alternative approach within LBM framework. By directly tracking the evolution of macroscopic fluid properties, the physical boundary conditions can be implemented in SHSLBM straightforwardly. For Dirichlet boundary conditions, the variables on boundary nodes are directly set to given values. For Neumann boundary conditions, boundary values of macroscopic variables are determined from the values of several layers of inner meshes to ensure desired order of accuracy.

However, it is noteworthy that, by using original formulations, boundary conditions for non-equilibrium distribution functions are still required (at least on non-uniform mesh) since their boundary values are needed in the corrector step [21]. The non-equilibrium extrapolation scheme was adopted to determine boundary values of the non-equilibrium distribution functions [25]. Such boundary treatment is based on the recognition that the field of non-equilibrium distribution function is smooth enough for extrapolation, which is a mathematical assumption rather than a physical interpretation. The revised scheme of SHSLBM, on the contrary, is free from such concerns. By adopting Eq. (8) rather than Eq. (4) in the corrector step, only the boundary values of equilibrium distribution function are needed. These boundary values can be analytically obtained by substituting boundary values of macroscopic variables into Eq. (6). Since no extrapolation scheme is adopted, this strategy

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can be regarded as an analytical approach to implement physical boundary conditions, which is physically more robust than the original scheme of SHSLBM.

One should also be reminded that the overall accuracy of the method is also constrained by the flow solver. Since SHSLBM itself is in the second-order of accuracy in space, the non-equilibrium extrapolation scheme [25] matches the order of accuracy of the flow solver. Therefore, using the revised scheme of SHSLBM cannot improve the order of accuracy. Even though, smaller numerical error can still be expected with the enhanced physical robustness.

2.4 Computational sequence

The computational sequence of the SHSLBM for the isothermal incompressible flow can be summarized as:

(1) Specify the streaming distance δ_x ($\delta_x = \delta_t$). Determine the single relaxation parameter τ.

(2) Predictor step: Use Eqs. (10) and (11) to calculate the intermediate physical properties ρ^* and $\mathbf{u}^*$.

(3) Determine boundary values of ρ^* and $\mathbf{u}^*$ from physical boundary conditions.

(4) Corrector step: Obtain the density and the velocity at the new time step from Eqs. (12) and (13).

(5) Implement appropriate boundary conditions for the macroscopic physical properties.

(6) Repeat steps (2)-(5) until the computation converges or the prescribed maximum computational time is reached.

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3 Stability Analysis

As its name implies, SHSLBM has very nice performance in numerical stability. In previous work, the scheme was analyzed through von Neumann stability analysis [21, 26]. However, certain technical flaws exist in those analyses, and the claim of "unconditionally stable" is not convincing. In this section, we offer the corrected stability analysis of SHSLBM.

To begin with, the following variables are defined

$$y_1 = \rho, \quad y_2 = \rho u, \quad y_3 = \rho v \tag{14}$$

And the vector of variables can be written as

$$\mathbf{Y} = \begin{pmatrix} y_1 & y_2 & y_3 \end{pmatrix}^{\mathrm{T}} \tag{15}$$

Substituting the above notations into the equilibrium distribution function (Eq. (6)) yields:

$$f_\alpha^{eq} = w_\alpha y_{1\alpha} + \frac{w_\alpha}{c_s^2}\left(e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha}\right) + \frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha}\right)^2}{y_{1\alpha}} - \frac{w_\alpha}{2c_s^2}\frac{y_{2\alpha}^2 + y_{3\alpha}^2}{y_{1\alpha}} \tag{16}$$

where $y_{1\alpha}$, $y_{2\alpha}$, and $y_{3\alpha}$ denote the functional values of y_1, y_2 and y_3 along a specific lattice direction, i.e., at the space location ($\mathbf{r}$-$\mathbf{e}_\alpha\delta_t$), respectively. Eq. (16) can be substituted into the expressions for intermediate density and moments (Eqs. (10) and (11)) to obtain a set of nonlinear algebraic equations, denoted as $\mathbf{Y}^*=(y_1^*, y_2^*, y_3^*)^T=F(\mathbf{Y}^n)$. However, the von Neumann stability analysis is only applicable to linear systems. To linearize the obtained equations, we differentiate them with respect to y_1, y_2, and y_3; and the resultant equations can be written as

$$\overline{\mathrm{d}\mathbf{Y}}^* = \mathbf{E}\cdot\overline{\mathrm{d}\mathbf{Y}}^n \tag{17}$$

where

$$E_{ij} = \left(\frac{\partial y_i^*}{\partial y_j}\right)^n \tag{18}$$

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and $\overline{\mathrm{d}y}_i^*$ and $\overline{\mathrm{d}y}_i^n$ are the magnitudes of $\mathrm{d}y_i^* = \overline{\mathrm{d}y}_i^* e^{j\mathbf{k}\cdot\mathbf{r}}$ and $\mathrm{d}y_i^n = \overline{\mathrm{d}y}_i^n e^{j\mathbf{k}\cdot\mathbf{r}}$ in the von Neumann stability analysis, respectively. The detailed derivation of the above relationship will be presented in the Appendix.

The formulations in the corrector step (Eqs.(12) and(13)) can also be linearized and analyzed following similar procedures to obtain a final formulation for stability analysis

$$\overline{\mathbf{dy}}^{n+1} = \mathbf{G}\cdot\overline{\mathbf{dy}}^{n} \tag{19}$$

where **G** is a the characteristic matrix with 3×3 dimensions. The detailed process is included in the Appendix.

For D2Q9 model, the resultant characteristic matrix **G**, as well as its maximum eigenvalue $\lambda_{\max}$, are functions of ρ, u, v, θ_x, θ_y and τ, i.e.,

$$\lambda_{\max} = f\left(\rho, u, v, \theta_x, \theta_y, \tau\right) \tag{20}$$

where $\theta_x=k_x\delta_t$, $\theta_y=k_y\delta_t$.

Following Sterling & Chen's suggestion [5], analysis is carried out at the "most unstable condition", i.e., the velocity vector and the wavenumber vector are both aligned with the horizontal direction. Meanwhile, the "most unstable wavenumber" was also introduced in Sterling & Chen's work [5] to further simplify the analysis. The "most unstable wavenumber" is determined by setting τ=0.5 and a specific value of mean flow velocity |**u**| and sweeping wavenumber from 0 to 2π. Under this value of τ, numerical instability is most likely to happen. The maximum eigenvalues are depicted in Figure 2. Note that, in this figure, the results of conventional LBM are collected from Sterling & Chen's work [5] in which hexagonal lattice velocity model is adopted; and only the points with eigenvalues larger than 1 are presented. Several important information reveals in this figure. Firstly, at the same mean flow velocity, the present SHSLBM is more stable than the conventional LBM. When mean flow velocity is 0.2, maximum eigenvalues of the characteristic matrix of SHSLBM at

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all wavenumbers are smaller than or equal to 1, while some maximum eigenvalues of conventional LBM are larger than 1 which represents the likelihood of numerical instability. Secondly, considering various mean flow velocities, the numerical instability of SHSLBM is most likely to happen at $k=\pi$. Finally, it is interesting to see that, under the incompressible limit (Ma<0.3 or $|\mathbf{u}|<0.173$), SHSLBM is always stable at $\tau=0.5$. Such characteristic makes SHSLBM advantageous over conventional LBM which becomes more unstable when τ is approaching to 0.5.

Given the above "most unstable wavenumber", stability boundary can be given as a function of mean flow velocity and relaxation parameter τ by setting threshold eigenvalue as 1. Using typical D2Q9 model, the stability boundaries of SHSLBM and conventional LBM [5] are presented in Figure 3. Since the stability boundary of SHSLBM is on the right side of that of conventional LBM, it means that SHSLBM has a larger stability region and is more stable than conventional LBM.

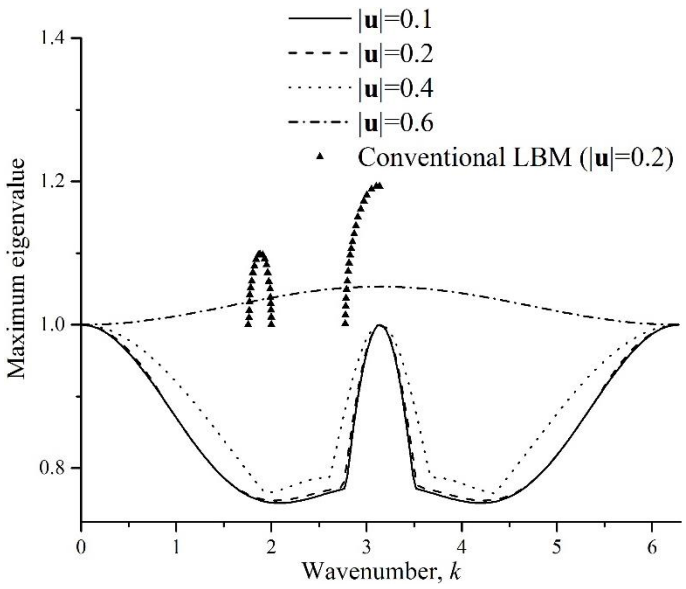


Figure 2 Maximum eigenvalue magnitude as function of wavenumber for τ=0.5.

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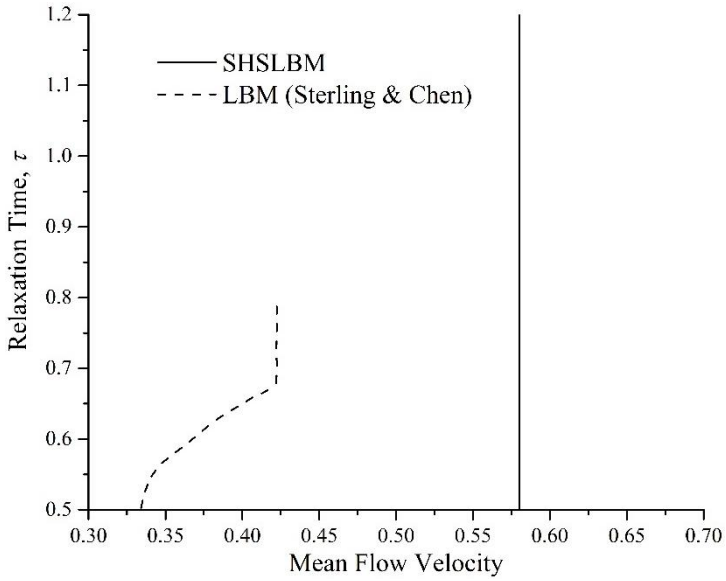


Figure 3 Stability boundary of D2Q9 model for SHSLBM and conventional LBM

4 Numerical Examples

To comprehensively assess the performance of SHSLBM, five benchmark tests are presented in this section, which include Taylor-Green vortex flow, plane Poiseuille flow, lid-driven cavity flow, flow past a stationary circular cylinder, and polar lid-driven cavity flow. Spatial accuracy, numerical convergence, stability, accuracy, and the flexibility on non-uniform meshes are studied in these examples to validate the robustness of SHSLBM.

4.1 Taylor-Green vortex flow

Taylor-Green vortex flow is a benchmark test for evaluating the practical order of accuracy of the numerical method. In this section, we firstly use this example to examine the spatial accuracy of SHSLBM, and compare its performance with SLBM. Theoretical solutions to Taylor-Green vortex flow can be written as

$$u(x, y, t) = -U_0 \cos(\pi x / L)\sin(\pi y / L)\exp\left(-\frac{2\pi^2 U_0 t}{\mathrm{Re}\, L}\right) \tag{21}$$

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$$v(x, y, t) = U_0 \sin(\pi x / L)\cos(\pi y / L)\exp\left(-\frac{2\pi^2 U_0 t}{\mathrm{Re}\, L}\right) \tag{22}$$

$$p(x, y, t) = p_0 - \frac{\rho_0 U_0^2}{4}\left[\cos(2\pi x / L) + \sin(2\pi y / L)\right]\exp\left(-\frac{4\pi^2 U_0 t}{\mathrm{Re}\, L}\right) \tag{23}$$

where U_0 and ρ_0 are the initial velocity amplitude and the reference density; and p_0 is the background pressure.

Five sets of mesh sizes (11×11, 21×21, 51×51 and 101×101) are adopted for numerical experiments. Reynolds number (Re=U_0L/υ) is chosen as 10. Solutions at the non-dimensional time $t^*=t(U_0/L)$=2.0 are extracted for analysis. And the L_2 norm of the relative error to the analytical solutions is defined as:

$$L_2 = \sqrt{\frac{\sum_{k=1}^{N\times N}\left(\frac{u_{\mathrm{numerical}} - u_{\mathrm{exact}}}{U_0}\right)^2}{N \times N}} \tag{24}$$

where $u_{\mathrm{numerical}}$ and u_{exact} refer to the numerical result and the analytical solution, respectively. Figure 4 presents the L_2 norms obtained by SHSLBM in log scale. It reveals that the slope of the linearly fitted line is 1.928, which indicates that the present method has approximately the second-order of accuracy in space.

Next, we utilize this example to validate the conclusion in stability analysis. The stability analysis reveals that SHSLBM is stable at τ=0.5 under the incompressible limit. Here, we set the characteristic velocity U_0=0.173 and τ=0.5 to perform numerical test. Instantaneous contours of horizontal velocity at dimensionless time T=1 and T=2 are shown in Figure 5. As can be seen in the figure, the computation of SHSLBM is stable at this extreme condition. And the magnitude of velocity generally remains the same as time marches, which is a typical inviscid characteristic.

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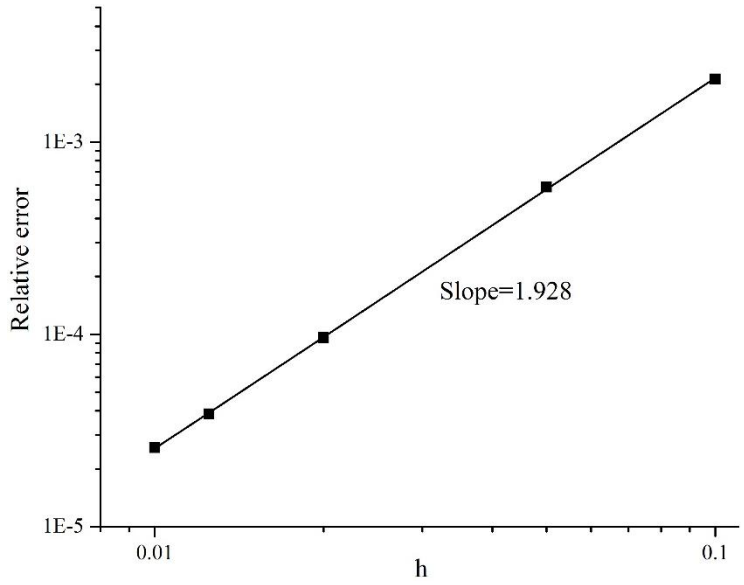


Figure 4 L_2 norms of relative error of u versus h in Taylor-Green vortex flow

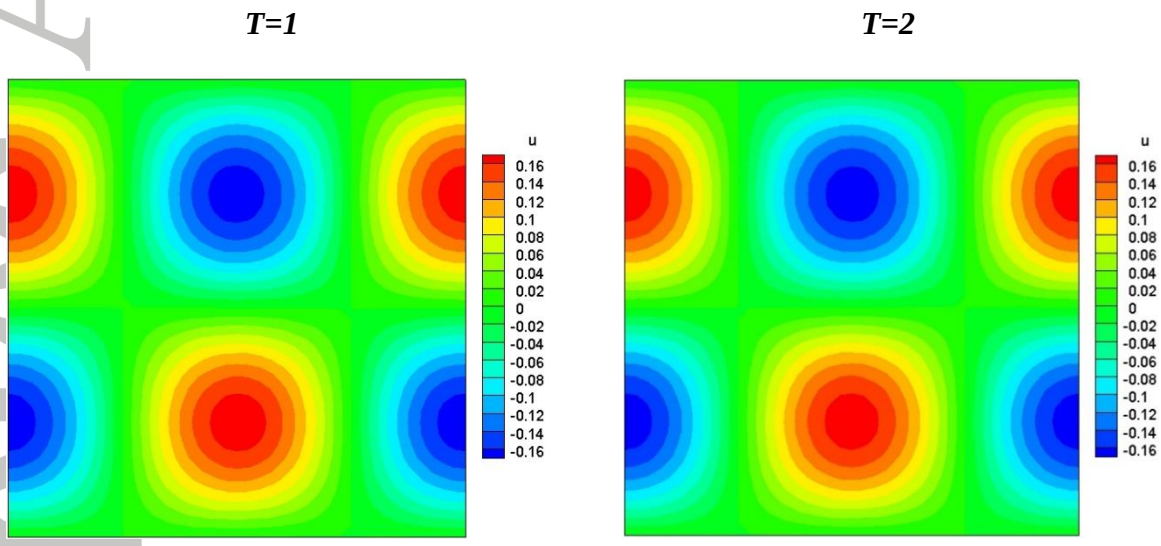


Figure 5 Instantaneous contours of horizontal velocity in Taylor-Green vortex flow for τ=0.5 and U_0=0.173

4.2 Plane Poiseuille flow

The plane Poiseuille flow is a classic benchmark test for incompressible viscous flows with analytical solutions available for comparisons. The physical configuration of the plane Poiseuille flow is the viscous flow between two parallel plates in distance of h=1. And the flow is driven by a uniform pressure gradient of $\partial p/\partial x$=-0.0001. The dynamic viscosity of the

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fluid is μ=0.001. From the Navier-Stokes equations, it can be shown that the analytical solution of the velocity profile for this problem is

$$u(y) = \frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)\left(y^2 - hy\right) \tag{25}$$

In our simulations, uniform mesh with the size of 21×25 is applied. No-slip boundary condition is implemented on the upper and the bottom walls, while uniform pressure boundary condition is applied on the left and the right boundaries.

Figure 6 shows the velocity distributions at the steady state. Here, the results obtained by the present scheme of SHSLBM and those given by the original scheme of SHSLBM with non-equilibrium extrapolation boundary treatment are compared with the analytical solutions. As can be seen, under the same mesh size, the revised scheme of SHSLBM gives more accurate result as compared to the original one. It is also noted that the discrepancy between these two results is quite small. This is caused by the overall constrain of the flow solver in spatial accuracy. Since the non-equilibrium extrapolation scheme also has the second-order of accuracy, its consolidation with the flow solver does not affect the overall order of accuracy. Even though, the comparison in Figure 6 indicates that smaller numerical error can still be expected by using the revised scheme of SHSLBM proposed in this paper.

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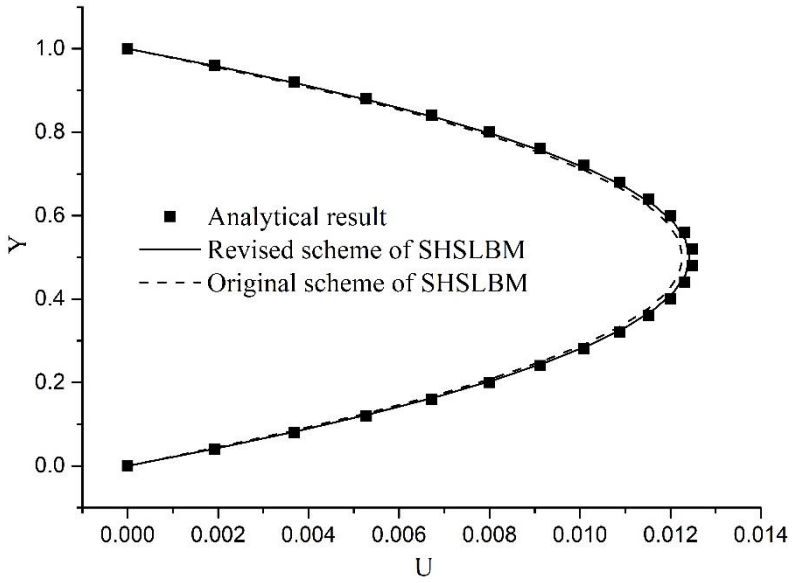


Figure 6 Vertical velocity distribution in plane Poiseuille flow

4.3 Lid-driven cavity flow

The example of plane Poiseuille flow can be regarded as a simple 1D problem with negligible velocity in the vertical direction. To comprehensively evaluate the performance of SHSLBM, 2D lid-driven cavity flow is chosen as the test case in this section. Apart from validating the numerical accuracy, we also use this example to demonstrate the convergence and the numerical stability of the present method. In this problem, the fluid inside a square cavity is driven by the lid moving at a constant velocity (denoted as U_{lid}) until reaching a steady circulating state. To fulfill the incompressible limit, we set U_{lid}=0.1. The Reynolds number, which is a critical parameter in determining the final flow pattern, is defined as $Re=U_{lid}L/\upsilon$, where L is the length of the square cavity.

To validate the convergence and the stability performance of the present method, we choose the case with Reynolds number of 7500 to perform tests. Various mesh sizes, including 11×11, 41×41, 81×81, 151×151, 251×251 and 401×401, are applied in simulations. As can be seen in Figure 7, the velocity profiles gradually converge to the reference data [27] by refining the mesh resolution. With the mesh sizes of larger than 251×251, the obtained

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velocity profiles generally overlap, which indicates that the computation has converged. Another notable issue is that at very coarse mesh resolution, e.g. 11×11, SHSLBM can still trigger the convergence criteria and give smooth velocity profile. Such performance demonstrates the nice numerical stability of the method, which is consistent to the theoretical analysis conducted in Section 3. For illustrative purposes, the time history of the relative error in SHSLBM simulation with mesh size of 151×151 is presented in Figure 8 and is compared with the results obtained by conventional LBM with regular non-equilibrium extrapolation scheme implemented on the boundaries. Specifically, the relative error is defined as

$$\varepsilon = \frac{\sum_{ij}\left|\left(\sqrt{u^2+v^2}\right)^{n+1} - \left(\sqrt{u^2+v^2}\right)^{n}\right|}{\sum_{ij}\left(\sqrt{u^2+v^2}\right)^{n+1}} \tag{26}$$

Apparently, conventional LBM cannot well converge at this mesh resolution and Reynolds number as its relative error is stabilized above 10^{-6} after certain iteration steps. On the contrary, the error of both the original and the present revised form of SHSLBM keeps decreasing as computation continues. This figure clearly shows the advantage of SHSLBM over conventional LBM in numerical stability. And the revised scheme and the original scheme of SHSLBM reveals similar rate of convergence.

Table 1 presents a comparative study of computational efficiency of the original and the revised schemes of SHSLBM. It should be noted that the computational efficiency is largely relative to programming tricks. Therefore, the comparative study presented here is only of reference value to readers. In our tests, two mesh sizes of 151×151 and 301×301 are considered; and the computation ceases at the iteration step of 5×10^5. In both cases, the CPU time of the revised scheme of SHSLBM is smaller than that of the original scheme of SHSLBM. The cases presented here suggest that roughly more than 30% of reduction in computational time can be expected.

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To further validate the accuracy of the revised scheme of SHSLBM, two cases with different Reynolds number are simulated and compared with the results obtained by the original scheme of SHSLBM with non-equilibrium extrapolation boundary condition. Mesh sizes of 251×251 and 301×301 are used in the cases with Re=7500 and 10000, respectively. As can be seen in Figure 9, in both cases, the converged velocity profiles obtained by the present SHSLBM agree well with the reference data reported by Ghia et al [27]. Compared with the original scheme of SHSLBM with non-equilibrium extrapolation boundary treatment, better agreement with the reference data is observed in the results given by the revised scheme of SHSLBM, which indicates the smaller numerical error in the revised scheme of SHSLBM.

Streamlines of the computational results at Re=7500 and Re=10000 are also presented in Figure 10. As can be seen, delicate flow structures (e.g. the secondary corner vortex at the right-bottom corner) are well captured by SHSLBM. Such performance proves that SHSLBM can preserve numerical accuracy while improving numerical stability.

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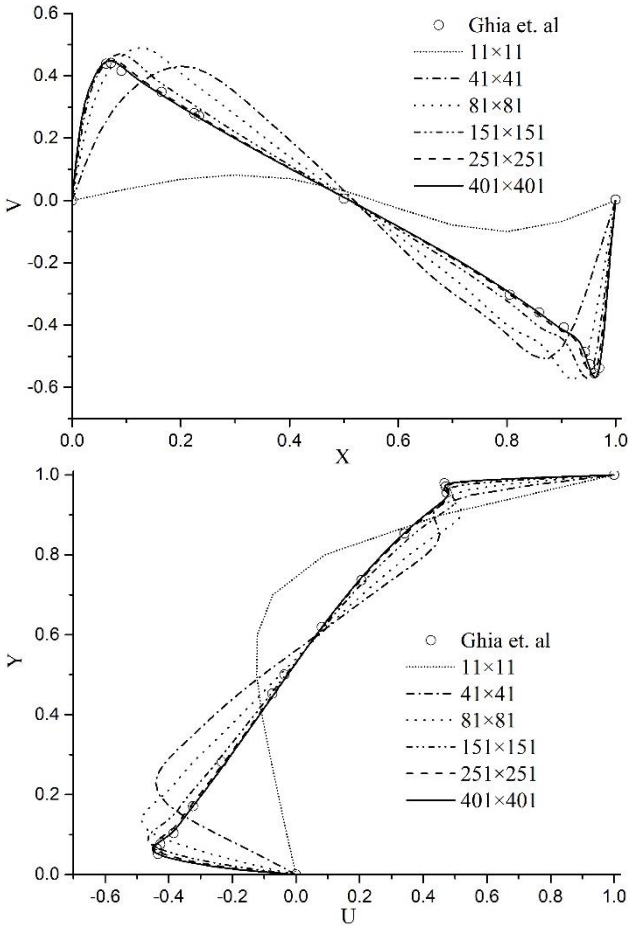


Figure 7 Velocity distributions along the centerlines of the lid-driven cavity at Re=7500

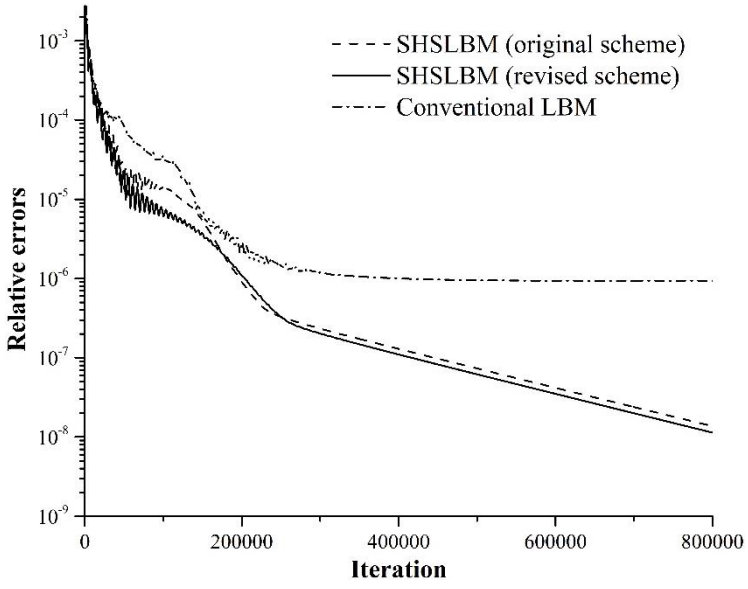
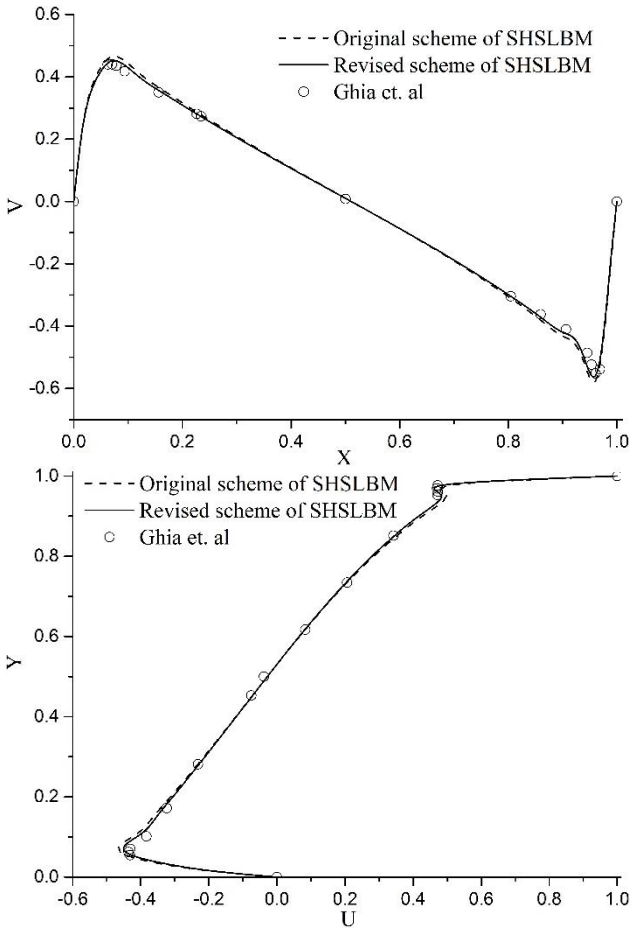


Figure 8 Time history of relative error of SHSLBM and conventional LBM in lid-driven cavity flow with Re=7500 and mesh size of 151×151.

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Table 1 Comparison of CPU time of the original the revised schemes of SHSLBM. Iteration steps: 5×10^5.

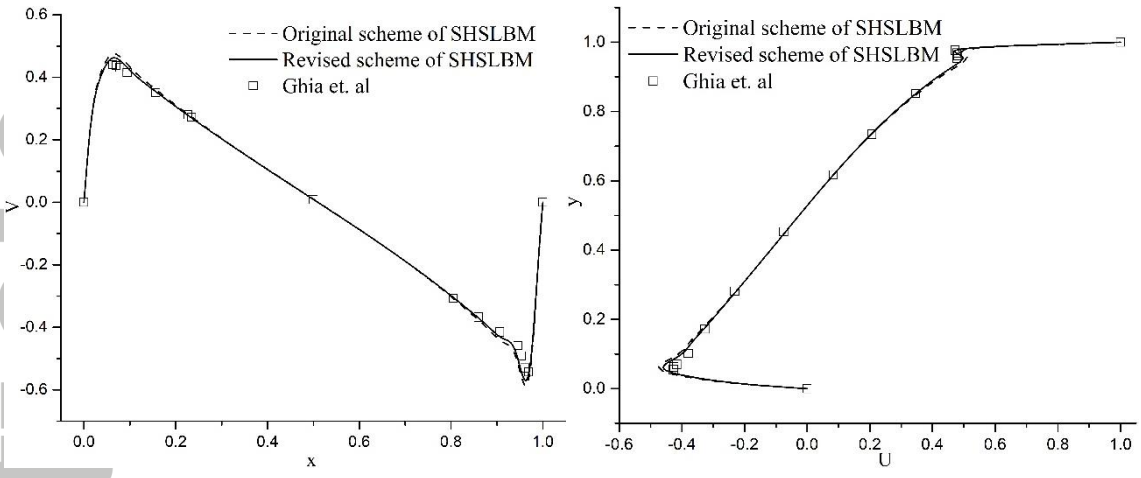
Mesh size	151×151	301×301
Original scheme of SHSLBM	4968 s	23316 s
Revised scheme of SHSLBM	3038 s	13243 s
Reduction rate	38.8%	43.3%



Re=7500

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Re=10000

Figure 9 Velocity distributions along the centerlines in lid-driven cavity flow. Left: x-v curve; Right u-y curve.

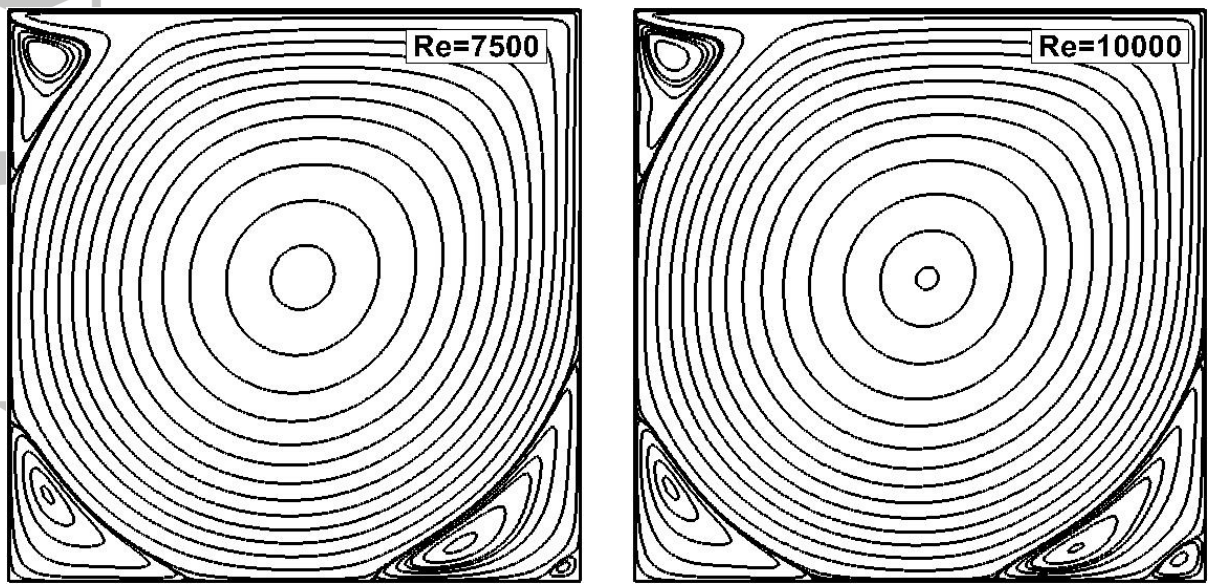


Figure 10 Streamlines of lid-driven cavity flow at Re=7500 and Re=10000.

4.4 Flow past a stationary circular cylinder

To evaluate the flexibility of SHSLBM on non-uniform meshes, the flow past a stationary circular cylinder is chosen as the test example. Physically, the simulation models the incoming viscous fluid with a constant free-stream velocity flowing over a fixed circular cylinder. To meet the incompressible limit, the free-stream velocity is chosen as U_0=0.1; and the Reynolds number, which affects the flow pattern, is defined as Re=$U_0 d/\upsilon$, where d is the

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diameter of the circular cylinder. Typical O-type mesh is applied in our simulations. Lagrange interpolation algorithm is introduced to evaluate the functional values at streaming locations which usually do not locate on the mesh points. No-slip boundary condition is implemented on the inner boundary, while free-stream condition is assigned on the outer boundary. Mesh sizes of 201×301 and 301×501 are used in the steady and unsteady cases, respectively. And the edges of the computational domains are 35*d* in steady cases and 75*d* in unsteady cases.

The drag and the lift coefficients are two key parameters, which are defined as

$$C_d = \frac{F_D}{\frac{1}{2}\rho U_{\inf}^2} \tag{27}$$

$$C_l = \frac{F_L}{\frac{1}{2}\rho U_{\inf}^2} \tag{28}$$

where F_D and F_L are the drag and the lift forces, respectively. They can be obtained by integrating the momentum equations over the surface of the cylinder, i.e.,

$$F_D = -\int_{\Gamma}\left[pn_x - \mu\frac{\partial u}{\partial \mathbf{n}}\right]\mathrm{d}\Gamma \tag{29}$$

$$F_L = -\int_{\Gamma}\left[-pn_y + \mu\frac{\partial v}{\partial \mathbf{n}}\right]\mathrm{d}\Gamma \tag{30}$$

where Γ represents the surface of the cylinder, n_x and n_y are the horizontal and the vertical components of the outer unit normal vector **n**, and μ is the dynamic viscosity of the fluid.

Depending on the Reynolds number, the flow can be classified as steady or unsteady. At low Reynolds number, the viscous effect is relatively strong; therefore, the recirculation zones maintain attached behind the cylinder and the flow is steady. Higher than certain values of Reynolds number, the viscous effect cannot ensure the stable attachment of the recirculation zones. Consequently, periodic vortex shedding is formed and the flow is called

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unsteady flow. For the steady flow, two cases with Reynolds numbers of 20 and 40 are considered here; and two other cases with Reynolds numbers of 100 and 200 are selected as unsteady cases. For quantitative comparisons, the drag coefficient C_d and the length of the recirculation zone l_s in steady cases are tabulated and compared with the reference data, as shown in Table 2; and the dynamic parameters of unsteady cases are presented in Table 3. As can be seen, the results of the original scheme and the revised scheme of SHSLBM are all within the range of reference data, which validates the accuracy and the flexibility on non-uniform meshes.

Table 2 Comparison of drag coefficient C_d and recirculation length l_s for steady flows at Re=20 & 40

Re	References	C_d	l_s/D
20	Dennis & Chang [28]	2.05	0.94
	Shukla et al. [29]	2.07	0.92
	Wu & Shu [30]	2.091	0.93
	SHSLBM (original scheme)	**2.086**	**0.92**
	SHSLBM (revised scheme)	**2.085**	**0.90**
40	Dennis & Chang [28]	1.52	2.35
	Shukla et al. [29]	1.55	2.34
	Wu & Shu [30]	1.565	2.31
	SHSLBM (original scheme)	**1.535**	**2.29**
	SHSLBM (revised scheme)	**1.529**	**2.30**

Table 3 Comparison of dynamic parameters of unsteady flow past a circular cylinder at Re=100 & 200

Re	References	C_l	C_d	St
100	Braza et al. [31]	±0.30	1.28 ±0.02	0.16
	Benson et al. [32]	±0.38	1.46 ±0.01	0.17
	Ding et al. [33]	±0.28	1.325 ±0.008	0.164

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	SHSLBM (original scheme)	**±0.33**	**1.340±0.009**	**0.162**
	SHSLBM (revised scheme)	**±0.34**	**1.331±0.012**	**0.162**
200	Braza et al. [31]	±0.78	1.38±0.07	0.19
	Benson et al. [32]	±0.65	1.45±0.04	0.193
	Ding et al. [33]	±0.60	1.327±0.045	0.196
	SHSLBM (original scheme)	**±0.69**	**1.338±0.045**	**0.195**
	SHSLBM (revised scheme)	**±0.72**	**1.348±0.046**	**0.193**

4.5 Polar lid-driven cavity flow

In this section, we test SHSLBM on flow in a polar cavity to further investigate its performance on non-uniform mesh and on problems with curved boundaries. The computational domain of this problem is a sector spanning the angle of θ=1. The inner boundary and outer boundary of the sector are denoted as R_{i} and R_{o}, respectively. And the flow inside is driven by the inner wall moving in a constant azimuthal velocity of U_θ until reaching a steady state. Reynolds number in this problem is defined as $\mathrm{Re}=U_\theta R_{\mathrm{i}}/\upsilon$, and its value is chosen as 350. Specifically, parameters used in our simulations are R_{i}=1.0, R_{o}=2.0, ρ_0=1.0 and U_θ=0.1. The mesh adopted in computation is uniform in the tangential direction and non-uniform in the radial direction. To capture high velocity gradient near inner and outer walls, mesh points along the radial direction are distributed as

$$r_i = 0.5\left[1-\chi\tan^{-1}\left(\left(1-\kappa_i\right)\tan\left(1/\chi\right)\right)\right] \tag{31}$$

$$\kappa_i = \frac{i-1}{2\left(i_{\max}-1\right)} \tag{32}$$

where χ=0.85 is the mesh-stretching parameter.

Mesh sizes of 41×41 and 81×81 are used in the computation. The converged streamlines are depicted in Figure 11. Apparently, SHSLBM maintains the smoothness of the flow field

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and successfully captures two minor vortices at right-top and right-bottom corners. Figure 12 offers the quantitative comparison of the velocity distribution along the horizontal line of θ=0.5. The obtained results are compared with reference data reported in literatures [34, 35]. Apart from showing its good agreement with the reference data, it is also noteworthy that the revised scheme of SHSLBM can give converged results at the coarser mesh size of 41×41, while the original scheme of SHSLBM requires finer mesh resolution. This is caused by the nonlinear correlation between the computational coordinate and the physical coordinate. As mentioned in Section 2.2, the revised scheme of SHSLBM can get rid of the shifting of the interpolation points, which accurately ensures the overall second-order of accuracy and induces smaller numerical error at the same mesh size.

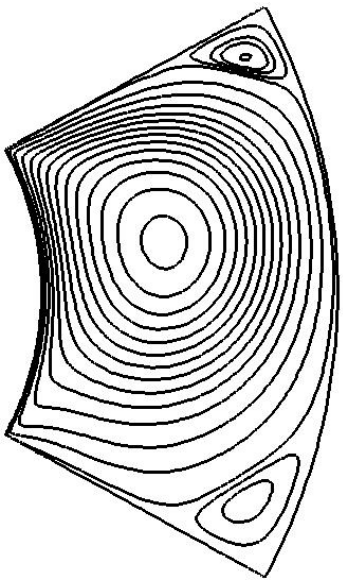


Figure 11 Streamlines of the polar lid-driven cavity flow at Re=350

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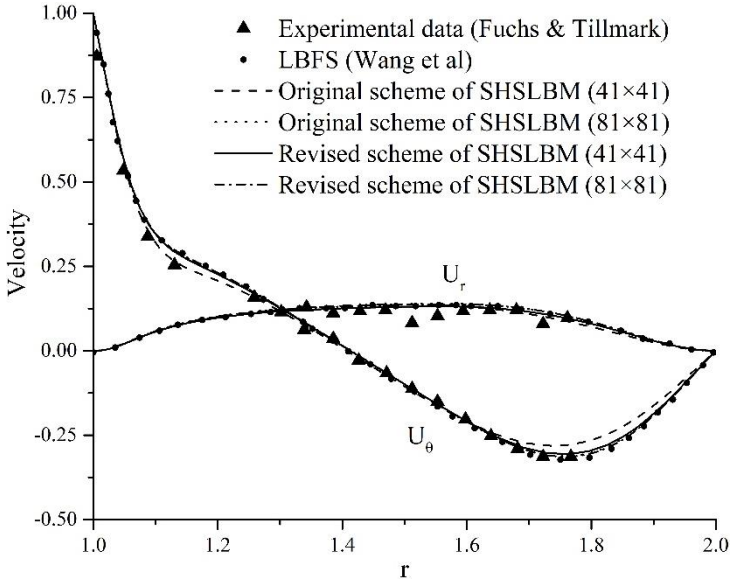


Figure 12 Distributions of radial and azimuthal velocities along the horizontal line of θ=0.5 for the polar lid-driven cavity flow at Re=350

5 Concluding Remarks

In this paper, we present an improved form of the simplified and highly stable lattice Boltzmann method (SHSLBM) which ensures the second-order of accuracy both in space and in time. This method is developed by reconstructing solutions to the macroscopic governing equations recovered from lattice Boltzmann equation and resolved in a predictor-corrector scheme. SHSLBM directly updates the macroscopic variables rather than the distribution functions, which greatly saves virtual memories and facilitates the implementation of physical boundary conditions. As shown in the paper, the improved scheme is more suitable for computations on non-uniform body-fitted meshes. Meanwhile, no boundary treatment for the non-equilibrium distribution function is needed, which avoids the mathematical extrapolation introduced in previous works and is physically more robust.

As its name implies, the present SHSLBM is a very stable method, which can be theoretically proven through the classic and robust von Neumann stability analysis. Specifically, when the relaxation parameter τ is very close to or even equal to 0.5,

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eigenvalues of the characteristic matrix of SHSLBM at all wavenumbers are always smaller than or equal to 1 under the incompressible limit. Theoretical analysis is then further validated numerically through Taylor-Green vortex flow and lid-driven cavity flow. It is found in Taylor-Green vortex flow that SHSLBM can well simulate inviscid flow with little numerical diffusions. It is also revealed in lid-driven cavity flow that SHSLBM can still trigger the convergence criteria under severe conditions at very high Reynolds number and on very coarse meshes. Such stability performance gives SHSLBM much advantage over the conventional LBM which usually shows poor numerical instability at high Reynolds number, and makes it a promising tool to real engineering problems at high Reynolds numbers.

Five representative benchmark tests, including Taylor-Green vortex flow, plane Poiseuille flow, lid-driven cavity flow, flow past a stationary circular cylinder and flow in a polar cavity, are modelled by the proposed method for comprehensive evaluation of its robustness and flexibility. It is shown in the tests that, compared to the original scheme of SHSLBM, the improved scheme of SHSLBM generally gives more accurate results at same mesh sizes. By incorporating normal interpolation method, SHSLBM can also be extended to problems on non-uniform meshes; and the obtained numerical results are in good agreement with reference data reported in literatures.

Appendix

Von Neumann stability analysis of SHSLBM

In this section, we will show the details in the von Neumann stability analysis performed on SHSLBM. With the notations introduced in Eq. (14), the equilibrium distribution function (6)can be written in the form shown in Eq. (16). By substituting Eq. (16) into Eqs. (10) and(11), the intermediate density and moments can be expressed as

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$$y_1^* = \sum_\alpha \left[w_\alpha y_{1\alpha}^n + \frac{w_\alpha}{c_s^2}\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right) + \frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} \right] \quad (-33)$$

$$y_2^* = \sum_\alpha \left[w_\alpha e_{\alpha x} y_{1\alpha}^n + \frac{w_\alpha e_{\alpha x}}{c_s^2}\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right) + \frac{w_\alpha e_{\alpha x}}{2c_s^4}\frac{\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha e_{\alpha x}}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} \right] \quad (-34)$$

$$y_3^* = \sum_\alpha \left[w_\alpha e_{\alpha y} y_{1\alpha}^n + \frac{w_\alpha e_{\alpha y}}{c_s^2}\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right) + \frac{w_\alpha e_{\alpha y}}{2c_s^4}\frac{\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha e_{\alpha y}}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} \right] \quad (-35)$$

where the superscript *n* represents the time level.

To apply von-Neumann stability analysis, it is assumed that the variables $\mathrm{d}y_i^*, \mathrm{d}y_i^n, y_{i\alpha}^n$ can be approximated by the Fourier serial expansion [26]. Consider a random component of these variables, which can be written as

$$\mathrm{d}y_i^* = \overline{\mathrm{d}y}_i^* e^{j\mathbf{k}\cdot\mathbf{r}};\ \mathrm{d}y_i^n = \overline{\mathrm{d}y}_i^n e^{j\mathbf{k}\cdot\mathbf{r}};\ y_{i\alpha}^n = y_i^n e^{j\mathbf{k}\cdot\left(\mathbf{r}-\mathbf{e}_\alpha\delta_t\right)},\quad i=1,2,3 \quad (-36)$$

As stated in Section 3, the obtained algebraic equations are nonlinear. Therefore, to linearize the problem, we substitute Eq. (-36) into Eqs. (-33)-(-35) and differentiate them with respect to y_1, y_2, and y_3 to obtain a set of linear equations as indicated in Eq. (17) with parameters

$$E_{ij} = \left(\frac{\partial y_i^*}{\partial y_j}\right)^n \quad (-37)$$

For the purpose of illustration, the first term in Eq. (-37) is given explicitly as

$$\begin{aligned}\left(\frac{\partial y_1^*}{\partial y_1}\right)^n = &\sum_\alpha w_\alpha e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} - \frac{\left(y_2^n\right)^2}{2c_s^4\left(y_1^n\right)^2}\sum_\alpha w_\alpha e_{\alpha x}^2 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} - \frac{\left(y_3^n\right)^2}{2c_s^4\left(y_1^n\right)^2}\sum_\alpha w_\alpha e_{\alpha y}^2 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} \\ &- \frac{y_2^n y_3^n}{c_s^4\left(y_1^n\right)^2}\sum_\alpha w_\alpha e_{\alpha x} e_{\alpha y} e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} + \frac{\left(y_2^n\right)^2 + \left(y_3^n\right)^2}{2c_s^2\left(y_1^n\right)^2}\sum_\alpha w_\alpha e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t}\end{aligned} \quad (-38)$$

Also note that for D2Q9 model, we can derive the following relationships

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$$
\begin{aligned}
&\sum_\alpha w_\alpha e_{\alpha x} e_{\alpha y}^2 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = -i\left[\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]\\
&\sum_\alpha w_\alpha e_{\alpha x}^2 e_{\alpha y} e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = -i\left[\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]\\
&\sum_\alpha w_\alpha e_{\alpha x}^3 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = -i\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]\\
&\sum_\alpha w_\alpha e_{\alpha y}^3 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = -i\left[\frac{2}{9}\sin(\theta_y)+\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]\\
&\sum_\alpha w_\alpha e_{\alpha y}^2 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = \frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)\\
&\sum_\alpha w_\alpha e_{\alpha x}^2 e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = \frac{2}{9}\cos(\theta_x)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)\\
&\sum_\alpha w_\alpha e_{\alpha x} e_{\alpha y} e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = 0\\
&\sum_\alpha w_\alpha e_{\alpha y} e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = -i\left[\frac{2}{9}\sin(\theta_y)+\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]\\
&\sum_\alpha w_\alpha e_{\alpha x} e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = -i\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]\\
&\sum_\alpha w_\alpha e^{-i\mathbf{k}\cdot\mathbf{e}_\alpha\delta_t} = \frac{4}{9}+\frac{2}{9}\cos(\theta_x)+\frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)
\end{aligned}
\qquad (-39)
$$

where $\theta_x=k_x\delta_t$, $\theta_y=k_y\delta_t$. With these relationships, Eqs. (-33)-(-35) and (-38) can be respectively simplified as

$$
\begin{aligned}
y_1^* = {} & y_1^n\left[\frac{4}{9}+\frac{2}{9}\cos(\theta_x)+\frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)\right]\\
& -iy_2^n\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]-iy_3^n\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right]\\
& +\frac{(y_2^n)^2}{y_1^n}\left[-\frac{2}{3}+\frac{2}{3}\cos(\theta_x)-\frac{1}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]\\
& +\frac{(y_3^n)^2}{y_1^n}\left[-\frac{2}{3}-\frac{1}{3}\cos(\theta_x)+\frac{2}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]
\end{aligned}
\qquad (-40)
$$

$$
\begin{aligned}
y_2^* = {} & -iy_1^n\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]+y_2^n\left[\frac{2}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]\\
& -i\frac{(y_2^n)^2}{y_1^n}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]-i\frac{(y_3^n)^2}{y_1^n}\left[-\frac{1}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]\\
& -i\frac{y_2^n y_3^n}{y_1^n}\left[\cos(\theta_x)\sin(\theta_y)\right]
\end{aligned}
\qquad (-41)
$$

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$$y_3^* = -iy_1^n\left[\frac{2}{9}\sin(\theta_y)+\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]+y_3^n\left[\frac{2}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]$$
$$-i\frac{(y_2^n)^2}{y_1^n}\left[-\frac{1}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right]-i\frac{(y_3^n)^2}{y_1^n}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right] \quad (-42)$$
$$-i\frac{y_2^n y_3^n}{y_1^n}\left[\sin(\theta_x)\cos(\theta_y)\right]$$

$$E_{11}=\left(\frac{\partial y_1^*}{\partial y_1}\right)^n=\frac{4}{9}+\frac{2}{9}\cos(\theta_x)+\frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)$$
$$+\frac{(y_2^n)^2}{(y_1^n)^2}\left[\frac{2}{3}-\frac{2}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_y)-\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right] \quad (-43)$$
$$+\frac{(y_3^n)^2}{(y_1^n)^2}\left[\frac{2}{3}+\frac{1}{3}\cos(\theta_x)-\frac{2}{3}\cos(\theta_y)-\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]$$

Through similar processes, other terms in (-37) can be derived:

$$E_{12}=-i\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]$$
$$+\frac{y_2^n}{y_1^n}\left[-\frac{4}{3}+\frac{4}{3}\cos(\theta_x)-\frac{2}{3}\cos(\theta_y)+\frac{2}{3}\cos(\theta_x)\cos(\theta_y)\right] \quad (-44)$$

$$E_{13}=-i\left[\frac{2}{9}\sin(\theta_y)+\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]$$
$$+\frac{y_3^n}{y_1^n}\left[-\frac{4}{3}-\frac{2}{3}\cos(\theta_x)+\frac{4}{3}\cos(\theta_y)+\frac{2}{3}\cos(\theta_x)\cos(\theta_y)\right] \quad (-45)$$

$$E_{21}=-i\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]+\frac{(y_2^n)^2}{(y_1^n)^2}i\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]$$
$$+\frac{(y_3^n)^2}{(y_1^n)^2}i\left[-\frac{1}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]+\frac{y_2^n y_3^n}{(y_1^n)^2}i\left[\cos(\theta_x)\sin(\theta_y)\right] \quad (-46)$$

$$E_{22}=\left[\frac{2}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]+\frac{y_2^n}{y_1^n}i\left[-\frac{4}{3}\sin(\theta_x)-\frac{2}{3}\sin(\theta_x)\cos(\theta_y)\right]$$
$$-\frac{y_3^n}{y_1^n}i\left[\cos(\theta_x)\sin(\theta_y)\right] \quad (-47)$$

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$$E_{23} = -\frac{y_2^n}{y_1^n} i\left[\cos(\theta_x)\sin(\theta_y)\right] + \frac{y_3^n}{y_1^n} i\left[\frac{2}{3}\sin(\theta_x) - \frac{2}{3}\sin(\theta_x)\cos(\theta_y)\right] \quad (-48)$$

$$\begin{aligned} E_{31} = &-i\left[\frac{2}{9}\sin(\theta_y) + \frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right] + \frac{(y_2^n)^2}{(y_1^n)^2} i\left[-\frac{2}{3}\sin(\theta_y) + \frac{1}{6}\cos(\theta_x)\sin(\theta_y)\right] \\ &+ \frac{(y_3^n)^2}{(y_1^n)^2} i\left[\frac{1}{3}\sin(\theta_y) + \frac{1}{6}\cos(\theta_x)\sin(\theta_y)\right] + \frac{y_2^n y_3^n}{(y_1^n)^2} i\left[\sin(\theta_x)\cos(\theta_y)\right] \end{aligned} \quad (-49)$$

$$E_{32} = \frac{y_2^n}{y_1^n} i\left[\frac{2}{3}\sin(\theta_y) - \frac{2}{3}\cos(\theta_x)\sin(\theta_y)\right] - \frac{y_3^n}{y_1^n} i\left[\sin(\theta_x)\cos(\theta_y)\right] \quad (-50)$$

$$E_{33} = \frac{y_2^n}{y_1^n} i\left[\frac{2}{3}\sin(\theta_y) - \frac{2}{3}\cos(\theta_x)\sin(\theta_y)\right] - \frac{y_3^n}{y_1^n} i\left[\sin(\theta_x)\cos(\theta_y)\right] \quad (-51)$$

In the corrector step, the formulations can be written as

$$y_i^{n+1} = y_i^* + (\tau - 1)\left(S_i - y_i^n\right) \quad (-52)$$

where

$$\begin{aligned} S_1 &= y_1^n \\ S_2 &= \sum_\alpha e_{\alpha x} f_\alpha^{eq*}\left(\mathbf{r} + \mathbf{e}_\alpha \delta_t, t\right) \\ S_3 &= \sum_\alpha e_{\alpha y} f_\alpha^{eq*}\left(\mathbf{r} + \mathbf{e}_\alpha \delta_t, t\right) \end{aligned} \quad (-53)$$

f_α^{eq*} is evaluated from the intermediate variables y_i^*.

By defining $\mathrm{d}y_i^{n+1} = \overline{\mathrm{d}y}_i^{n+1} e^{j\mathbf{k}\cdot\mathbf{r}}$, and following procedures shown in Eqs. (-36)-(-43), we can derive the following relationship for the corrector step:

$$\overline{\mathbf{dy}}^{n+1} = \overline{\mathbf{dy}}^* + \mathbf{F}\cdot\overline{\mathbf{dy}}^* + \mathbf{H}\cdot\overline{\mathbf{dy}}^n \quad (-54)$$

where

$$F_{ij} = (\tau - 1)\left(\frac{\partial S_i}{\partial y_j}\right)^* \quad (-55)$$

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$$\left(\frac{\partial S_1}{\partial y_1}\right)^* = 0 \quad \left(\frac{\partial S_1}{\partial y_2}\right)^* = 0 \quad \left(\frac{\partial S_1}{\partial y_3}\right)^* = 0 \tag{-56}$$

$$\begin{aligned}\left(\frac{\partial S_2}{\partial y_1}\right)^* &= i\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right]-\frac{(y_2^*)^2}{(y_1^*)^2}i\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]\\ &-\frac{(y_3^*)^2}{(y_1^*)^2}i\left[-\frac{1}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]-\frac{y_2^*y_3^*}{(y_1^*)^2}i\left[\cos(\theta_x)\sin(\theta_y)\right]\end{aligned} \tag{-57}$$

$$\begin{aligned}\left(\frac{\partial S_2}{\partial y_2}\right)^* &= \left[\frac{2}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right]-\frac{y_2^*}{y_1^*}i\left[-\frac{4}{3}\sin(\theta_x)-\frac{2}{3}\sin(\theta_x)\cos(\theta_y)\right]\\ &+\frac{y_3^*}{y_1^*}i\left[\cos(\theta_x)\sin(\theta_y)\right]\end{aligned} \tag{-58}$$

$$\left(\frac{\partial S_2}{\partial y_3}\right)^* = \frac{y_2^*}{y_1^*}i\left[\cos(\theta_x)\sin(\theta_y)\right]-\frac{y_3^*}{y_1^*}i\left[\frac{2}{3}\sin(\theta_x)-\frac{2}{3}\sin(\theta_x)\cos(\theta_y)\right] \tag{-59}$$

$$\begin{aligned}\left(\frac{\partial S_3}{\partial y_1}\right)^* &= i\left[\frac{2}{9}\sin(\theta_y)+\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]-\frac{(y_2^*)^2}{(y_1^*)^2}i\left[-\frac{2}{3}\sin(\theta_y)+\frac{1}{6}\cos(\theta_x)\sin(\theta_y)\right]\\ &-\frac{(y_3^*)^2}{(y_1^*)^2}i\left[\frac{1}{3}\sin(\theta_y)+\frac{1}{6}\cos(\theta_x)\sin(\theta_y)\right]-\frac{y_2^*y_3^*}{(y_1^*)^2}i\left[\sin(\theta_x)\cos(\theta_y)\right]\end{aligned} \tag{-60}$$

$$\left(\frac{\partial S_3}{\partial y_2}\right)^* = -\frac{y_2^*}{y_1^*}i\left[\frac{2}{3}\sin(\theta_y)-\frac{2}{3}\cos(\theta_x)\sin(\theta_y)\right]+\frac{y_3^*}{y_1^*}i\left[\sin(\theta_x)\cos(\theta_y)\right] \tag{-61}$$

$$\begin{aligned}\left(\frac{\partial S_3}{\partial y_3}\right)^* &= \frac{2}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)+\frac{y_2^*}{y_1^*}i\left[\sin(\theta_x)\cos(\theta_y)\right]\\ &-\frac{y_3^*}{y_1^*}i\left[-\frac{4}{3}\sin(\theta_y)-\frac{2}{3}\cos(\theta_x)\sin(\theta_y)\right]\end{aligned} \tag{-62}$$

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -(\tau-1) & 0 \\ 0 & 0 & -(\tau-1) \end{bmatrix} \tag{-63}$$

Together with the characteristic matrix **E** obtained in the predictor step and the formulations of y_i^* in Eqs. (-40)-(-42), we can derive

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$$\overline{\mathbf{dy}}^{n+1} = \left[\left[\mathbf{I}+\mathbf{F}\right]\cdot\mathbf{E}+\mathbf{H}\right]\cdot\overline{\mathbf{dy}}^{n} = \mathbf{G}\cdot\overline{\mathbf{dy}}^{n} \qquad (-64)$$

where G denotes the overall characteristic matrix for SHSLBM, which is then used for stability analysis.

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